Experiment

1. Open the Binomial module in your web application.

opened binomial module

2. Enter n = 20 and p = 0.05.

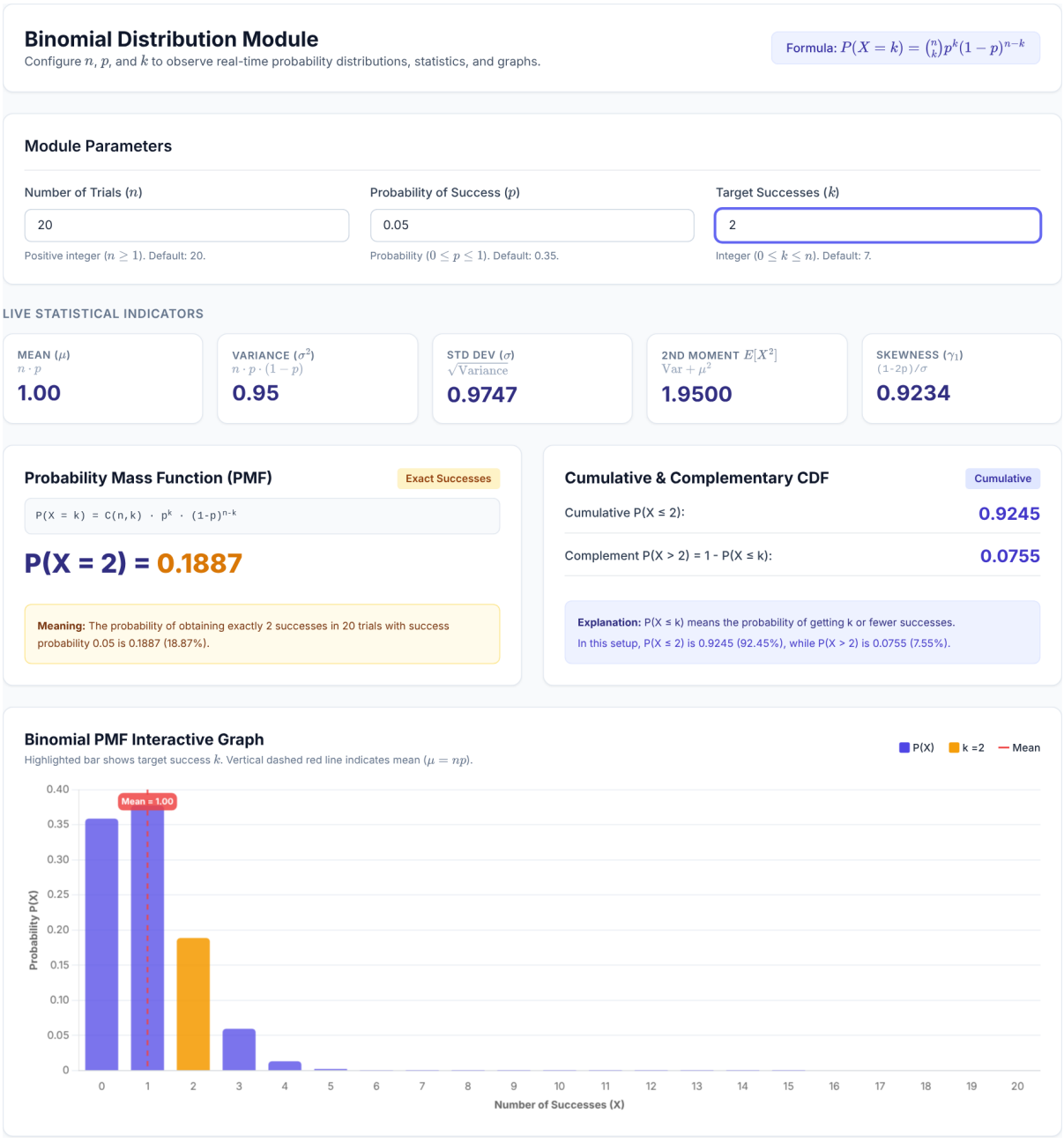
entered n = 20 and p = 0.05

3. Calculate the mean, variance, and probability of exactly 2 defects.

mean = np = 20 x 0.05 = 1

variance = np(1-p) = 20 x 0.05 x 0.95 = 0.95

P(X = 2) = 0.1887



binomial module with n = 20, p = 0.05, k = 2

4. View the PMF graph. The horizontal axis represents the number of defects and the vertical axis represents the probability of each count.

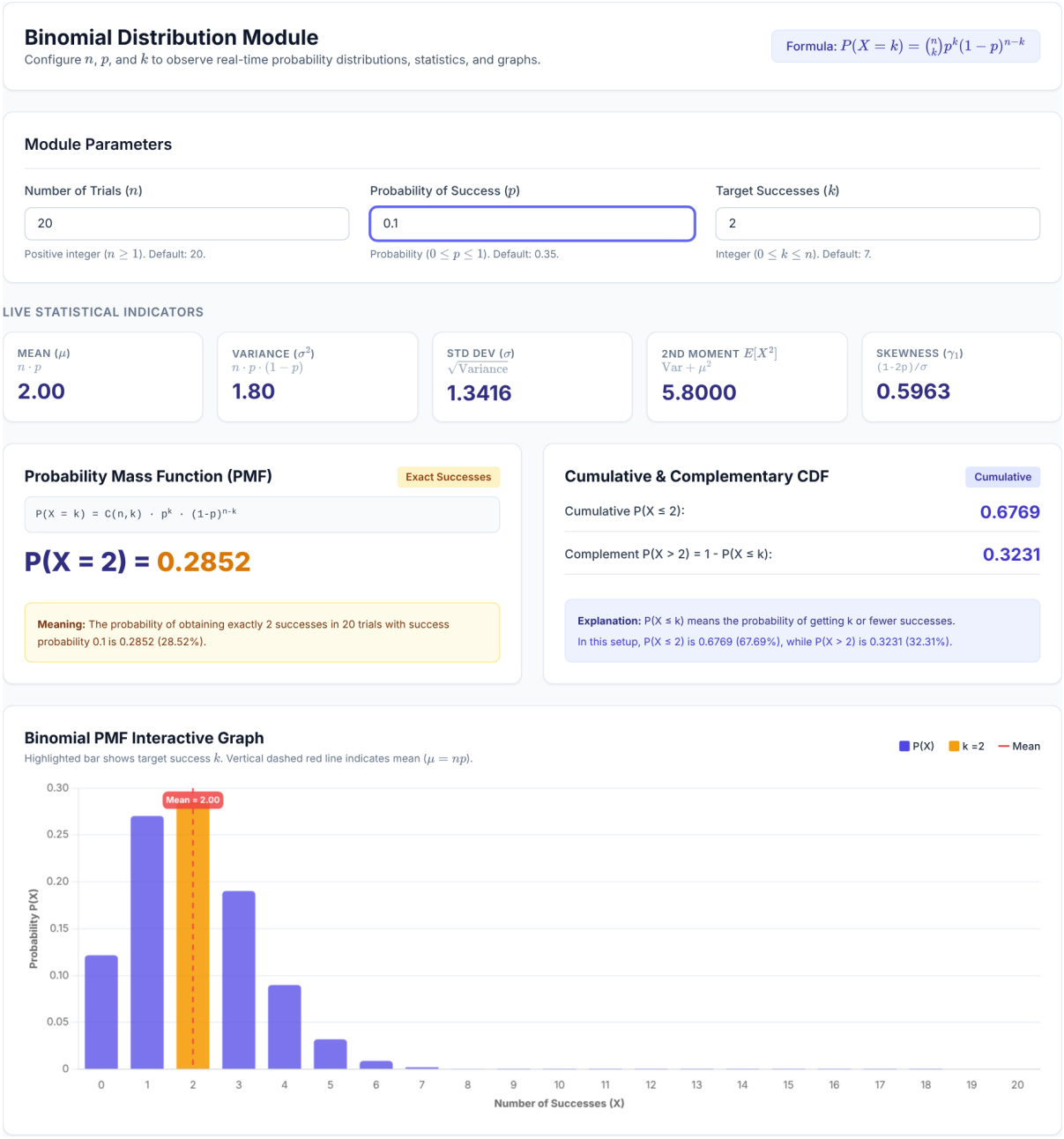
highest bar at 1 defect (0.3774), then 0 defects (0.3585) and 2 defects (0.1887)

after 4 defects the bars are almost zero so graph is right skewed (see graph in screenshot above)

5. Change the defect probability from 0.05 to 0.10. Observe how the distribution shifts toward larger defect counts.

with p = 0.10 mean = 2, variance = 1.80, P(X = 2) = 0.2852

peak moved from 1 to 2 defects so graph shifted right and got wider



binomial module with n = 20, p = 0.10, k = 2

6. Open the Poisson module and enter λ = 5 for 100 inspected products. Compare the Poisson probabilities with the Binomial results.

poisson λ = 5 : P(0) = 0.0067, P(2) = 0.0842, P(5) = 0.1755

binomial n = 100, p = 0.05 : P(0) = 0.0059, P(2) = 0.0812, P(5) = 0.1800

values are almost same so poisson can be used instead of binomial when n is big and p is small

Poisson Distribution Module

Set the average rate λ and a target count k to explore event-count probabilities in a fixed interval.

Formula: $P(X = k) = \frac{e^{-\lambda}\lambda^k}{k!}$

Module Parameters

Average Rate (λ)

5

Positive number ($0 < \lambda \leq 100$). Default: 4.

Target Events (k)

5

Non-negative integer ($k \geq 0$). Default: 3.

LIVE STATISTICAL INDICATORS

MEAN (μ)	VARIANCE (σ^2)	STD DEV (σ)	2ND MOMENT $E[X^2]$	SKEWNESS (γ_1)
λ	λ	$\sqrt{\lambda}$	$\lambda + \lambda^2$	$1 / \sqrt{\lambda}$
5.0000	5.0000	2.2361	30.0000	0.4472

Probability Mass Function (PMF)

Exact

$P(X = k) = e^{-\lambda} \cdot \lambda^k / k!$

P(X = 5) = 0.1755

Meaning: The probability of observing exactly 5 events when the average rate is λ = 5 is 0.1755 (17.55%).

Cumulative & Complementary CDF

Cumulative

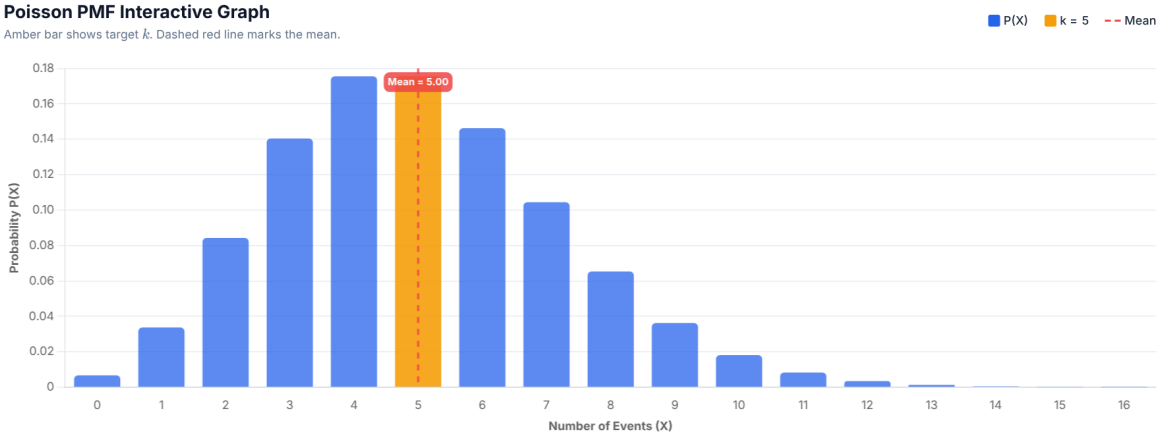
$P(X \leq k) = \Sigma_{i=0..k}\, e^{-\lambda}\, \lambda^i / i!$

P(X ≤ 5):	0.6160
P(X > 5) = 1 - P(X ≤ k):	0.3840

Explanation: P(X ≤ k) means the probability of observing k or fewer events in the interval. With λ = 5, P(X ≤ 5) is 0.6160 (61.60%), while P(X > 5) is 0.3840 (38.40%).

Poisson PMF Interactive Graph

Amber bar shows target k. Dashed red line marks the mean.



AIML Lab Context:

Poisson models counts of rare, independent events: requests hitting a server per second, defects per batch, or clicks per minute. In anomaly detection, an observed count with a very small tail probability $P(X > k)$ is flagged as unusual.

poisson module with λ = 5, k = 5

Probability Mass Function (PMF)

Exact

$P(X = k) = e^{-\lambda} \cdot \lambda^k / k!$

P(X = 0) = 0.0067

Meaning: The probability of observing exactly 0 events when the average rate is λ = 5 is 0.0067 (0.67%).

Cumulative & Complementary CDF

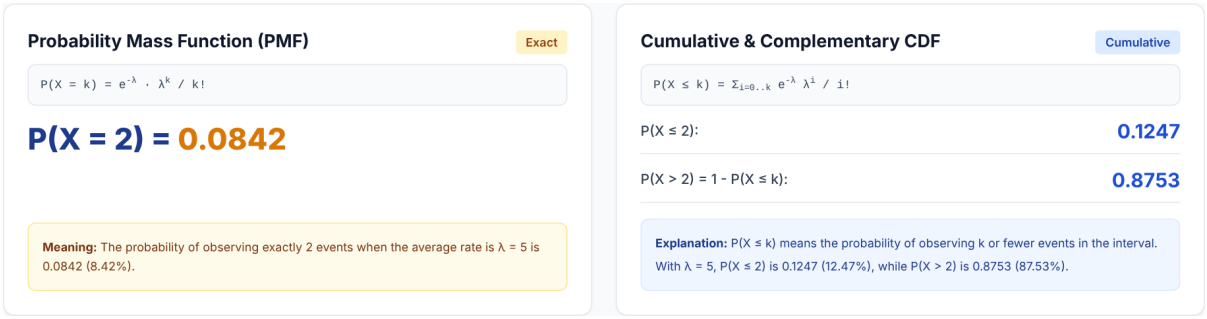
Cumulative

$P(X \leq k) = \Sigma_{i=0..k}\, e^{-\lambda}\, \lambda^i / i!$

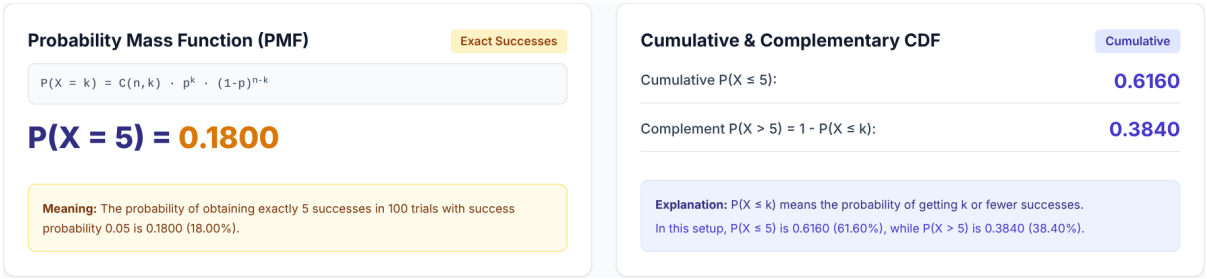
P(X ≤ 0):	0.0067
P(X > 0) = 1 - P(X ≤ k):	0.9933

Explanation: P(X ≤ k) means the probability of observing k or fewer events in the interval. With λ = 5, P(X ≤ 0) is 0.0067 (0.67%), while P(X > 0) is 0.9933 (99.33%).

poisson λ = 5, k = 0



poisson λ = 5, k = 2



binomial n = 100, p = 0.05, k = 5 for comparison

7. Open the Geometric module and enter p = 0.05. Examine the probability of detecting the first defect on different inspection attempts.

1st attempt = 0.05, 2nd = 0.0475, 3rd = 0.0451, 5th = 0.0407, 10th = 0.0315, 20th = 0.0189

mean = 1/p = 20 inspections

probability keeps decreasing slowly with every attempt

Geometric Distribution Module

Set the success probability p and a trial number k to explore how long it takes to get the first success.

Formula: $P(X = k) = (1 - p)^{k-1} p$

Module Parameters

Probability of Success (p)

0.05

Probability ($0 < p \leq 1$). Default: 0.2.

Trial of First Success (k)

1

Positive integer ($k \geq 1$). Default: 3.

LIVE STATISTICAL INDICATORS

MEAN (μ)	VARIANCE (σ^2)	STD DEV (σ)	2ND MOMENT $E[X^2]$	SKEWNESS (γ_1)
1 / p	(1 - p) / p²	√Variance	(2 - p) / p²	(2 - p) / √(1 - p)
20.0000	380.0000	19.4936	780.0000	2.0007

Probability Mass Function (PMF)

Exact

$P(X = k) = (1 - p)^{k-1} \cdot p$

P(X = 1) = 0.0500

Meaning: The probability that the first success happens exactly on trial 1 (after 0 failures) with p = 0.05 is 0.0500 (5.00%).

Cumulative & Complementary CDF

Cumulative

$P(X \leq k) = 1 - (1 - p)^k$

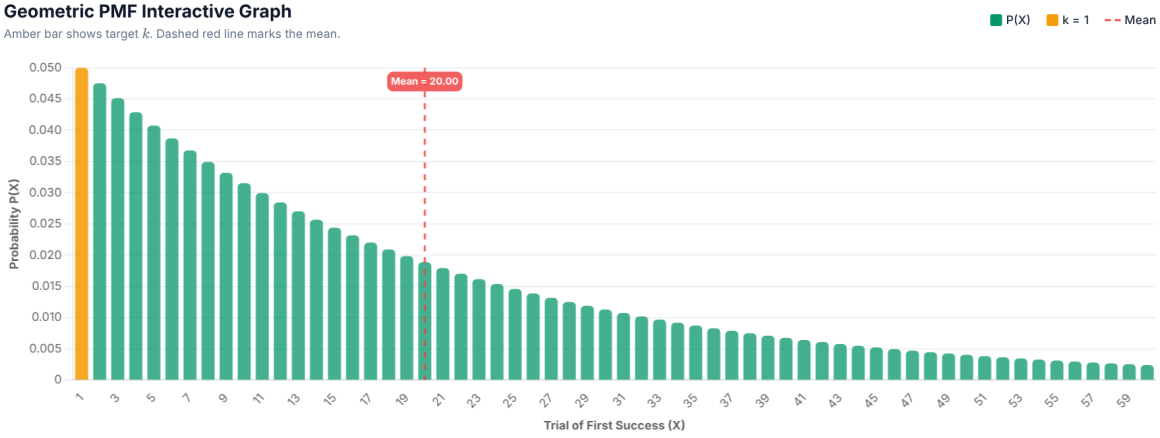
P(X ≤ 1): **0.0500**

P(X > 1) = 1 - P(X ≤ k): **0.9500**

Explanation: P(X ≤ k) means the probability that the first success happens within the first k trials.
With p = 0.05, the chance of succeeding within the first 1 trial is 0.0500 (5.00%); needing more than 1 trials is 0.9500 (95.00%).

Geometric PMF Interactive Graph

Amber bar shows target k. Dashed red line marks the mean.



AIML Lab Context:
Geometric models waiting time until the first success: retries until a network request succeeds, episodes until an RL agent first reaches the goal, or samples until a rare class appears. It is memoryless: $P(X > m + n \mid X > m) = P(X > n)$.

geometric module with p = 0.05, k = 1

Probability Mass Function (PMF)

Exact

$P(X = k) = (1 - p)^{k-1} \cdot p$

P(X = 2) = 0.0475

Meaning: The probability that the first success happens exactly on trial 2 (after 1 failure) with p = 0.05 is 0.0475 (4.75%).

Cumulative & Complementary CDF

Cumulative

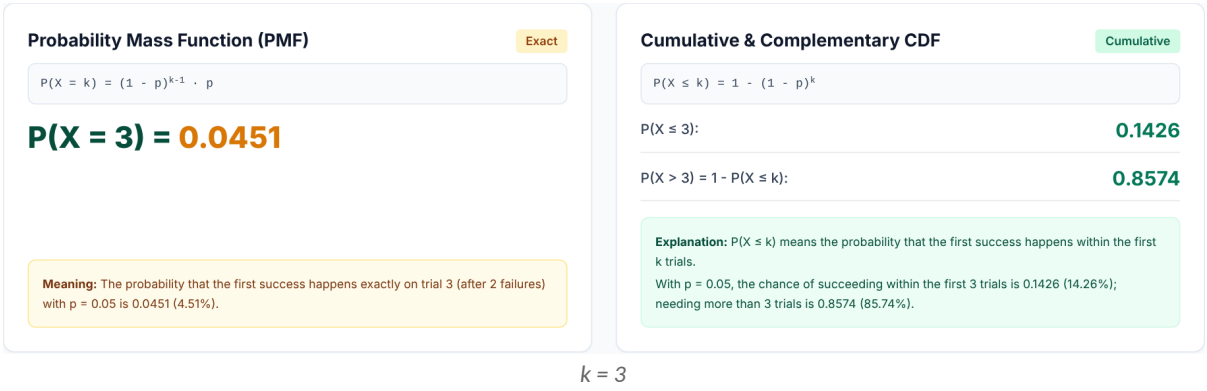
$P(X \leq k) = 1 - (1 - p)^k$

P(X ≤ 2): **0.0975**

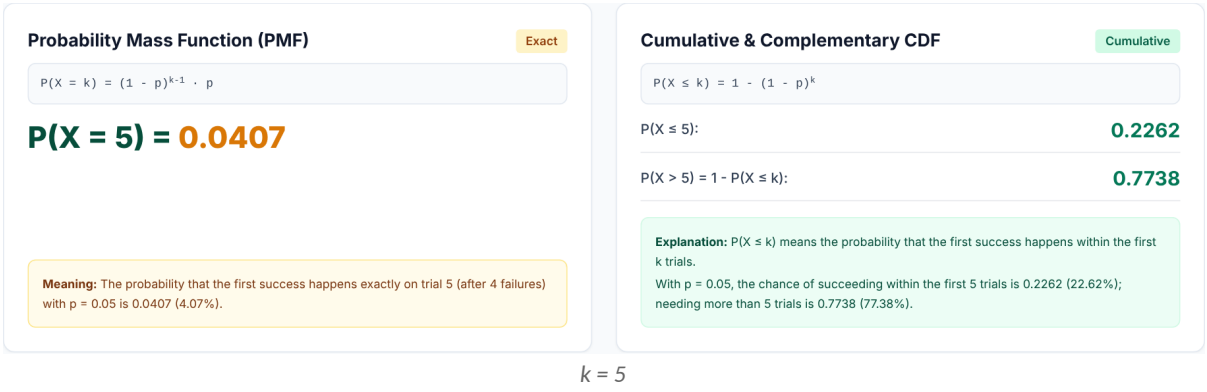
P(X > 2) = 1 - P(X ≤ k): **0.9025**

Explanation: P(X ≤ k) means the probability that the first success happens within the first k trials.
With p = 0.05, the chance of succeeding within the first 2 trials is 0.0975 (9.75%); needing more than 2 trials is 0.9025 (90.25%).

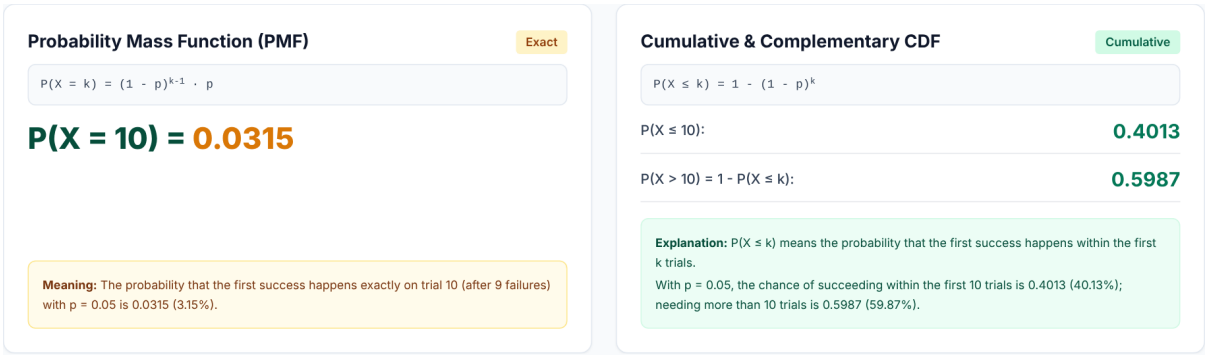
k = 2



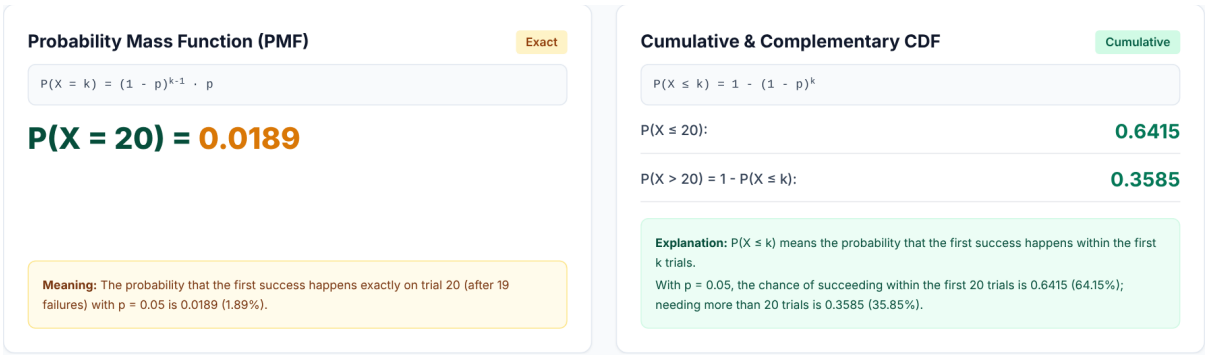
k = 3



k = 5



k = 10



k = 20

8. Write down the numerical results, graph observations, and an inference about the quality-control process.

results: p = 0.05 gives mean 1, variance 0.95, P(X = 2) = 0.1887 and p = 0.10 gives mean 2, variance 1.80, P(X = 2) = 0.2852

graphs: binomial graph shifts right when p increases, poisson graph looks almost same as binomial, geometric graph keeps going down

inference: at 5% defect rate the process is under control (mostly 0 to 2 defects in 20), at 10% more defects come so the process needs checking, poisson works for big batches and on avg 20 inspections are needed to find the first defect